

Frequency-Optimised Adversarial Patches for Semantic Segmentation:

What Has Been Measured, and What to Run Next

Findings, formal statement, and experimental design

Abstract

This document collects what has been established since the patch family moved from pixel-space realism constraints to a constraint stated in the *frequency domain* by a model of human contrast sensitivity, and turns it into guidance for the remaining experiments. Three results reframe the premise. First, the segmentation network is approximately *spectrally flat*: it reads every radial band up to Nyquist about equally, so the exploitable asymmetry is not a band that the network reads and the eye cannot see, but a $\sim 47\times$ difference in what a unit of prediction change *costs* perceptually. Second, the perceptual budget in use was mis-calibrated by a factor of ≈ 1.9 at the frequencies that matter, for two independent and partly cancelling reasons that are derived in closed form in Sec. 4.4. Third, a constraint that was believed to hold by construction was in fact violated by $2.8\text{--}7.4\times$ during optimisation, because clipping against real image content is a hard nonlinearity that generates broadband harmonics; enforcing it costs 4.6 mIoU points and is now mandatory for any number quoted with a τ attached. Section 8 states the three untargeted objectives — cross-entropy, CosPGD, and a Tsallis-entropy objective — with full derivations, including the closed form of the Tsallis gradient and its Shannon limit. Every symbol in every equation is defined where it first appears. Elements without an external source are marked [no external source] and tabulated in Sec. 12.

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1 Scope: what changed, and why this document exists

The patch family in this project has three generations. The first parameterised a patch in *pixel space* and constrained nothing (Brown et al., 2017). The second added *pixel-space realism*: a pull toward a reference image, a total-variation penalty, and printability, after Tan et al. (2021) and Sharif et al. (2016). Neither generation knows anything about an observer — both penalise statistics of the patch rather than its *visibility*.

The third generation, which this document is about, states the constraint in the frequency domain against a contrast sensitivity function (CSF). That converts “make the patch look plausible” into a falsifiable statement: a perturbation is admissible if its contrast at every spatial frequency sits below the detection threshold of a modelled observer at that frequency (Campbell and Robson, 1968; Barten, 2003).

The motivating asymmetry was stated as a hypothesis and has now been measured. It was: a segmentation network has no CSF and weights the band up to Nyquist uniformly, whereas the human visual system does not, so a perturbation may be admissible to one and not to the other. The named risk was that SegFormer’s patch embedding has stride 4 (Xie et al., 2021), so the highest frequencies might be attenuated before the first attention block ever sees them — in which case the attack would be bounded by an architectural detail rather than by the perceptual budget.

The premise survives, but not in its original form. Section 10.1 gives the measurement. The network reads *everything*, including the high band; what differs across bands is not effectiveness but perceptual price. The claim to make in the thesis is therefore about *efficiency*, and never about a band the network can see and the eye cannot.

Relation to the companion document. `docs/csf_patch_procedure.tex` states the procedure formally and carries its own provenance table. This document does not repeat those derivations except where a measurement has changed them; where the two disagree, this one is later.

How to read the findings. Findings are labelled **F1–F9** in Sec. 10 and are referred to by label from the design guidance in Sec. 11. Each states the measurement, the mechanism, and — most importantly — the claim it does *not* support.

2 Notation

Every symbol used anywhere below, in one place.

Images and the network

$\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$	input image, ImageNet-normalised, as the network receives it
$\mathbf{x}^{01}$	the same image in $[0, 1]$ sRGB <i>code</i> values, $\text{clip}(\mathbf{x} \odot \boldsymbol{\sigma}_n + \boldsymbol{\mu}_n, 0, 1)$, where $\boldsymbol{\mu}_n, \boldsymbol{\sigma}_n$ are the per-channel ImageNet statistics
H, W	image height and width in pixels; 512×1024 throughout unless stated
$\mathbf{y}$	ground-truth label map with trainIds in $\{0, \dots, C - 1\} \cup \{255\}$, where 255 is <i>void</i> and is never scored

C	number of classes; $C = 19$ for Cityscapes (Cordts et al., 2016)
f	the frozen segmentation network; $f(\mathbf{x}) \in \mathbb{R}^{C \times H' \times W'}$ are logits at the head's own resolution
$\mathcal{U}$	bilinear upsampling from $H' \times W'$ to $H \times W$
$\mathbf{p}_{uv}$	$= \text{softmax}(\mathcal{U}f(\cdot)_{:,u,v})$, the predicted class distribution at pixel (u, v) , a point in the simplex Δ^{C-1}
$\mathbf{e}_k$	the one-hot vector of class k in $\mathbb{R}^C$
The patch	
s	patch scale as a fraction of image height; $s = 0.25$
$p = \lfloor Hs \rfloor$	rendered patch side in pixels; $p = 128$ at $H = 512$, $s = 0.25$
S	parameter-grid resolution (<code>-patch_size</code>). Must equal p in frequency modes (Sec. 6.4)
$\mathbf{z} \in \mathbb{R}^{3 \times S \times S}$	the optimised parameter, unconstrained
$\boldsymbol{\delta} \in \mathbb{R}^{3 \times S \times S}$	the residual: a <i>signed</i> additive perturbation in sRGB code units
$\mathbf{r}$	the residual's <i>base</i> : the image content the patch covers (<code>-from_image</code>), or mid grey
$\mathcal{M}$	silhouette mask, 1 on pixels actually composited; all ones for a square patch
$\mathbf{m}$	the footprint: $\mathcal{M}$ padded into the full frame
Frequency	
$\mathcal{F}, \mathcal{F}^{-1}$	real 2-D DFT (<code>rfft2</code>) and its inverse, <i>forward-normalised</i> : $\mathcal{F}$ carries the $1/(S^2)$ factor, so a cosine of peak amplitude A has coefficient magnitude $A/2$
$\mathcal{K}$	the half-spectrum bin index set of an $S \times S$ real signal; $ \mathcal{K} = S(\lfloor S/2 \rfloor + 1)$
f_y, f_x	bin frequencies in cycles per pixel, each in $[-\frac{1}{2}, \frac{1}{2}]$
$\rho = \sqrt{f_y^2 + f_x^2}$	radial frequency in cycles per pixel (cyc/px); $\rho_{\max} = 1/2$ is Nyquist
θ	degrees of visual angle subtended by one pixel (Sec. 3.1)
$u = \rho/\theta$	radial frequency in cycles per degree (cpd) — the unit a CSF is defined in
The perceptual constraint	
$\text{CSF}(u)$	contrast sensitivity: the reciprocal of the just-detectable Michelson contrast at u
κ	contrast scale: converts a DFT magnitude into a Michelson contrast (Sec. 4.1)
$B(f)$	per-bin amplitude budget (Sec. 4.2)
$A_{\max}$	hard cap on B where $\text{CSF} \rightarrow 0$; $A_{\max} = 0.25$
β	Minkowski exponent modelling probability summation; $\beta = 3$
$V_\beta(\boldsymbol{\delta})$	visibility index of $\boldsymbol{\delta}$, in nominal JND (Sec. 5.2)
τ	the perceptual budget: the permitted value of V_β . The single knob of this attack family
Y	linear relative luminance in $[0, 1]$; Y_{ref} is the reference the CSF is evaluated at
c	an sRGB code value in $[0, 1]$; c_{ref} is the code value whose luminance is Y_{ref}
L	absolute luminance in cd/m^2 ; $L = Y L_{\text{peak}}$ with $L_{\text{peak}} = 100$
E	retinal illuminance in trolands (Td)
Evaluation	
Ω	the scored support: valid, non-footprint, optionally reach-restricted pixels (Sec. 8.1)
$\Delta \text{mIoU}_{\text{rem}}$	remote mIoU drop, the headline attack metric (Sec. 9.1)
ϕ	any-flip rate: fraction of remote pixels whose arg max changed (Sec. 9.2)

3 The observer model

A perceptual constraint is only as meaningful as the observer it assumes. This section fixes that observer completely: the geometry, the sensitivity function, and the transfer function relating the numbers the optimiser touches to the light a human would receive.

3.1 Viewing geometry: the assumption everything is conditional on

A CSF is defined over cycles per *degree of visual angle*. An image spectrum is in cycles per *pixel*. Converting between them requires a physical pixel size and a viewing distance:

$$\theta = \frac{180}{\pi} \cdot 2 \arctan\left(\frac{P}{2D}\right) \quad [\text{degrees per pixel}], \quad (1)$$

$$u = \rho/\theta \quad [\text{cycles per degree}]. \quad (2)$$

Here P is the physical size of one displayed pixel in cm, D is the distance from observer to display in cm, ρ is radial frequency in cyc/px from Sec. 2, and u is the frequency the CSF is evaluated at. The factor $180/\pi$ converts radians to degrees; the factor 2 with the half-angle $P/2D$ is the standard small-object subtense.

Only the ratio P/D matters. Equation (1) depends on P and D only through $P/(2D)$, so they are not two independent knobs but one. A geometry can therefore be quoted as a single number, θ in deg/px, and any (P, D) pair realising it is equivalent. Report θ ; never report P and D as though they were separate degrees of freedom.

Defaults, and why they are not the threat model. The defaults are $P = 0.0114$ cm and $D = 50$ cm, reproducing the geometry of Galinska et al. (2026, App. D), i.e. a laptop screen. They give $\theta \approx 0.01306$ deg/px and a Nyquist frequency of

$$u_{\text{Nyq}} = \frac{0.5}{\theta} \approx 38.3 \text{ cpd}, \quad (3)$$

which is *below* the ≈ 50 – 60 cpd where human sensitivity actually vanishes. Consequence, and it must be stated: the sampling grid cannot represent a truly invisible frequency. The budget at Nyquist is large but finite, and “invisible” here always means “below threshold”, never “outside the passband”.

Two constructors, because deriving P by hand is where the mistake happens. For a real display of diagonal d_{in} inches, N pixels across, and aspect ratio $a:b$,

$$P = \frac{2.54 d_{\text{in}} a}{N \sqrt{a^2 + b^2}}. \quad (4)$$

A 15.6-inch 1080p laptop gives $P = 0.0180$ cm, which is $1.6\times$ the default; a residual built for the default geometry and inspected on that laptop at 50 cm measures roughly $5\times$ its requested visibility. Nothing in the method is wrong there — the geometry is simply describing a different monitor.

For a *physical* patch of width w metres, viewed at distance d metres, rendered on an $S \times S$ grid,

$$\theta = \frac{1}{S} \cdot \frac{180}{\pi} \cdot 2 \arctan\left(\frac{w}{2d}\right). \quad (5)$$

A 0.5 m patch at 20 m on a 128 px grid gives $\theta = 0.01119$ deg/px, i.e. $u_{\text{Nyq}} = 44.7$ cpd: a *finer* geometry than the desk monitor. The road threat model therefore has *more* room to hide in than inspecting PNGs on a laptop suggests, which is the opposite of the intuition one gets from looking at saved images.

Validation by eye is a consistency check, not a perceptual study. A patch built for geometry θ may be inspected on any display of pixel density ppi provided it is viewed at

$$D_{\text{match}} = \frac{2.54/\text{ppi}}{2 \tan(\theta_{\text{rad}}/2)} \propto \frac{1}{\text{ppi}}, \quad (6)$$

with θ_{rad} the per-pixel angle in radians. At the default geometry that is 79 cm on a 141 PPI laptop and 119 cm on a 93 PPI desktop panel. Viewing closer is not a stricter test of the same claim; it is a test of a different claim.

3.2 Barten's CSF

The default sensitivity model is Barten's (2003) formula, with the parameter values of Galinska et al. (2026, App. A):

$$\text{CSF}(u) = \frac{M_{\text{opt}}(u)/k}{\sqrt{\frac{2}{T} \left(\frac{1}{X_0^2} + \frac{1}{X_{\text{max}}^2} + \frac{u^2}{N_{\text{max}}^2} \right) \left(\frac{1}{\eta_q p_\gamma E} + \frac{\Phi_0}{1 - e^{-(u/u_0)^2}} \right)}}. \quad (7)$$

symbol	value	meaning
$M_{\text{opt}}(u)$	—	optical modulation transfer of the eye, Eq. (8)
k	3.0	signal-to-noise ratio the visual system requires for detection
T	0.1 s	integration time of the eye
X_0	60°	angular size of the object
X_{max}	12°	maximum angular area the eye integrates over
N_{max}	15	maximum number of cycles integrated
η_q	0.03	quantum efficiency of the eye
p_γ	1.2×10^6	photon conversion factor, photons s ⁻¹ deg ⁻² Td ⁻¹
E	500 Td	retinal illuminance (revised in Sec. 4.4)
Φ_0	3×10^{-8} deg ² s	spectral density of neural noise
u_0	7 cpd	lateral-inhibition cutoff
σ	0.5/60 deg	standard deviation of the optical point spread

The optical transfer factor is

$$M_{\text{opt}}(u) = \exp\left(-2\pi^2 \sigma^2 u^2\right). \quad (8)$$

Reading the formula. The three factors have distinct roles. M_{opt} is the eye's optics and is a Gaussian low-pass: it is what kills sensitivity at high u . The first bracket under the root is *spatio-temporal integration* — how much signal the visual system can accumulate — and grows like u^2 , which is a second high-frequency penalty. The second bracket is *noise*: a photon-noise term $1/(\eta_q p_\gamma E)$ that shrinks as the scene gets brighter, plus a neural-noise term divided by a lateral-inhibition factor $1 - e^{-(u/u_0)^2}$ that vanishes at DC.

CSF(0) = 0, **and that is correct.** As $u \rightarrow 0$ the inhibition factor $1 - e^{-(u/u_0)^2} \rightarrow 0$, the neural-noise term diverges, and sensitivity goes to zero. Physically this says a uniform field has no contrast to detect. It is handled explicitly in code rather than being left to produce a NaN, because $1/\text{CSF}$ is what the budget of Sec. 4.2 is built from and a NaN there poisons everything downstream silently.

Sensitivity is a reciprocal threshold. $\text{CSF}(u) = 700$ means a grating of Michelson contrast $1/700$ at frequency u is just detectable. This is the definition that makes Sec. 4.2 an inversion rather than an analogy.

3.3 The Standard Spatial Observer, as a control

The alternative model is the Standard Spatial Observer (Watson and Ahumada, 2005; Watson, 2000), which unlike Barten is *orientation-dependent*:

$$\text{CSF}_{\text{SSO}}(f_x, f_y) = \text{CSF}_{\text{T}}(u) \cdot \text{OE}(f_x, f_y), \quad (9)$$

$$\text{CSF}_{\text{T}}(u) = g \left(e^{-u/f_m} - \ell e^{-u^2/s_o^2} \right), \quad (10)$$

$$\text{OE}(f_x, f_y) = 1 - w \frac{4 \left(1 - e^{-u/o_s} \right) f_x^2 f_y^2}{u^4}. \quad (11)$$

Here $u = \sqrt{f_x^2 + f_y^2}$ in cpd, $g = 330.74$ is an overall gain, $f_m = 7.28$ sets the exponential high-frequency decay, $\ell = 0.837$ is the low-frequency loss and $s_o = 1.809$ the scale at which that loss is attenuated, $w = 1$ weights the oblique effect ($w = 0$ disables it) and $o_s = 6.664$ is its scale.

The oblique effect is a free asymmetry. $\text{OE} = 1$ on the axes ($f_x = 0$ or $f_y = 0$) and dips to a minimum at 45° , where $f_x^2 f_y^2 / u^4 = 1/4$. Diagonal gratings are harder to see than axis-aligned ones of identical amplitude, so a diagonal carrier is cheaper perceptually at equal amplitude. This is not exploited by the current attack and is a legitimate future rung.

Why keep two models. So that the visibility claim can be shown not to depend on one particular CSF. If Barten and SSO disagree about a perturbation, that disagreement belongs in the write-up. SSO has a second use as a control: it has *no luminance parameter* (it fits a single photopic observer), so under SSO only the contrast denominator responds to background luminance and the two luminance effects of Sec. 4.4 are cleanly separated.

3.4 sRGB, luminance, and the unit bridge

The optimiser perturbs sRGB *code values*; contrast is defined on *linear luminance*. Conflating the two is the single easiest way to be wrong by a power of 2.4, and Sec. 4.4 shows it was worth a factor of 1.9 in the budget.

The sRGB transfer function and its inverse (International Electrotechnical Commission, 1999) are

$$Y(c) = \begin{cases} c/12.92, & c \leq 0.04045, \\ \left(\frac{c + 0.055}{1.055} \right)^{2.4}, & c > 0.04045, \end{cases} \quad (12)$$

$$c(Y) = \begin{cases} 12.92 Y, & Y \leq 0.0031308, \\ 1.055 Y^{1/2.4} - 0.055, & Y > 0.0031308, \end{cases} \quad (13)$$

with c the code value in $[0, 1]$ and Y the linear value in $[0, 1]$. The linear segment below 0.04045 is not decoration: a naive $c^{2.4}$ departs from the standard by tens of percent in exactly the dark codes that Cityscapes asphalt occupies.

The *local gain* of that curve is the unit bridge:

$$\left. \frac{dY}{dc} \right|_c = \begin{cases} 1/12.92, & c \leq 0.04045, \\ \frac{2.4}{1.055} \left(\frac{c + 0.055}{1.055} \right)^{1.4}, & c > 0.04045. \end{cases} \quad (14)$$

A budget derived in linear units is not a bound on the tensor the optimiser touches until it has been divided by this slope. The linearisation is first-order and therefore valid for small perturbations; at τ of order one the residual is far below one code step and this is comfortable, at large τ it is not, and there the *measured* realised visibility rather than the budget is the number to trust.

Relative luminance of a colour pixel uses the Rec. 709 weights (ITU-R, 2015), applied to *linearised* channels:

$$Y = 0.2126 Y(c_R) + 0.7152 Y(c_G) + 0.0722 Y(c_B). \quad (15)$$

Linearise *then* weight, never the reverse: averaging gamma-encoded channels over-states the luminance of a dark region, which would flatter every fixed-reference simplification in Sec. 10.3.

Finally, background luminance enters Barten's CSF through the pupil (Barten, 1999, Eq. 2.9):

$$d_{\text{pupil}}(L) = 5 - 3 \tanh(0.4 \log_{10} L) \quad [\text{mm}], \quad (16)$$

$$E(L) = \frac{\pi d_{\text{pupil}}(L)^2}{4} L \quad [\text{Td}], \quad (17)$$

with L the absolute background luminance in cd/m^2 . The pupil ranges from ≈ 8 mm in near-darkness to ≈ 2 mm in bright light. Equation (17) is the retinal illuminance E that appears in Eq. (7). This is the *entire* mechanism by which background luminance changes Barten's sensitivity, and it is why that dependence is comparatively flat: the pupil partly compensates.

4 The budget: inverting a sensitivity into an amplitude bound

4.1 From a DFT magnitude to a Michelson contrast

A DFT magnitude is not a contrast, and the gap is a constant that must be written down rather than absorbed silently. A real cosine of peak amplitude A riding on a mean level μ has Michelson contrast (Michelson, 1927)

$$c_M = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{(\mu + A) - (\mu - A)}{(\mu + A) + (\mu - A)} = \frac{A}{\mu}, \quad (18)$$

where $I_{\max}, I_{\min}$ are the maximum and minimum levels of the grating. With the forward-normalised transform of Sec. 2 that cosine has coefficient magnitude $|\mathcal{F}\delta(f)| = A/2$. Substituting,

$$c_M(f) = \kappa |\mathcal{F}\delta(f)|, \quad \kappa = \frac{2}{\mu}. \quad (19)$$

κ is the *contrast scale*. The legacy convention takes $\mu = \frac{1}{2}$ (mid grey), giving

$$\kappa_{\text{legacy}} = 4. \quad (20)$$

This convention is wrong on this dataset, and Sec. 4.4 says by how much. It is retained as the default *deliberately*, so that the existing τ ladder keeps its meaning and every past run is restatable by a constant factor rather than invalidated. Every results table must carry both a nominal and a calibrated column.

4.2 The amplitude budget

A component of contrast c at frequency u is at the detection threshold when $c \cdot \text{CSF}(u) = 1$, by the definition of sensitivity in Sec. 3.2. Requiring instead that each component reach at most τ just-noticeable differences and solving Eq. (19) for the magnitude gives the per-bin *amplitude budget* [**no external source**]

$$B(f) = \min\left(\frac{\tau}{\kappa \text{CSF}(u(f))}, A_{\max}\right), \quad (21)$$

where $f \in \mathcal{K}$ indexes a half-spectrum bin, $u(f)$ is that bin’s radial frequency in cpd via Eq. (2), τ is the perceptual budget, κ is from Eq. (19), and $A_{\max} = 0.25$ caps the budget where $\text{CSF} \rightarrow 0$ and $1/\text{CSF}$ would diverge. A quarter of the dynamic range in a single Fourier component is already far beyond anything this constraint is meant to permit, so the cap binds on no bin until $\tau \gtrsim 4$.

The shape of B is the whole premise. B is proportional to $1/\text{CSF}$, so it is small where the eye is sensitive (a few cpd) and large near Nyquist. That inversion is what “camouflage in the frequency domain” means operationally. The sanity check before trusting any run: if B at high frequency is not orders of magnitude larger than B at the CSF peak, there is no gap to hide in and the premise fails at step zero.

4.3 The low-frequency cutoff: making a full-field CSF apply to a patch

Equation (21) as written has a defect that inverts its intent. Since $\text{CSF}(0) = 0$, the bins nearest DC receive the *largest* allowance of any frequency. Measured: a residual built that way put 3×10^{-5} of its power in the lowest radial bin and 10^{-9} in the highest, and the diagnostic table printed a budget of 0.0770 for 0.00–0.02 cyc/px against 0.0026 at Nyquist — the exact opposite of what the attack is supposed to do.

This is not a bug in the CSF; it is a category error in applying it. The low-frequency roll-off is measured for *full-field* gratings, where the eye adapts to a slow luminance ramp. Inside a patch of side S there is no such grating: a component below $1/S$ cyc/px does not complete a single cycle across the patch, so it is not a grating but a brightness offset — and a patch whose mean differs from its surroundings is a visible rectangle whatever the CSF says about DC, because the patch border turns any offset into an edge, and edges are broadband.

The fix is a hard low-frequency cutoff [**no external source**]:

$$B(f) \leftarrow B(f) \cdot \mathbb{I}\left[\rho(f) \geq \frac{n_c}{S}\right], \quad n_c = 2, \quad (22)$$

where $\rho(f)$ is the bin’s radial frequency in cyc/px, S is the patch side in pixels, n_c is the minimum number of full cycles a component must complete across the patch, and $\mathbb{I}[\cdot]$ is the indicator function. Setting $n_c = 0$ restores the naive behaviour and is useful only for reproducing the failure. This cutoff carries over unchanged to every budget variant below; nothing about luminance or universality changes the argument for it.

4.4 Calibration: the budget was $\approx 1.9\times$ too permissive

Two of the constants above are wrong on Cityscapes, and they are wrong in opposite directions, so quoting only the net error hides the mechanism.

Error 1: the Michelson denominator. $\kappa = 4$ assumes $\mu = 0.5$. Measured over the full Cityscapes train split ($n = 2975$) with a 128 px footprint at 512×1024 , the median mean code value inside the footprint is 0.303 at centre placement and 0.292 on the road surface at (0.75, 0.5); the val split gives 0.311. In linear relative luminance the medians are $Y = 0.0971$, 0.0792 and 0.1029. So $\mu = 0.5$ is $1.65\times$ too bright at centre and $1.71\times$ on the road. Contrast is A/μ , so assuming a brighter background than the truth *under-states* contrast and over-states the admissible amplitude.

Error 2: the code-value/luminance conflation. $\kappa = 2/\mu$ implicitly treats code values as luminances. They are not. Redoing Eq. (19) correctly: a cosine of peak amplitude A_{code} in *code* units produces a luminance modulation $A_{\text{lin}} = (dY/dc) A_{\text{code}}$ to first order, and Michelson contrast is $A_{\text{lin}}/Y_{\text{ref}}$. Hence the *calibrated* contrast scale

$$\kappa_{\text{cal}}(Y_{\text{ref}}) = \frac{2}{Y_{\text{ref}}} \cdot \frac{dY}{dc} \Big|_{c_{\text{ref}}}, \quad c_{\text{ref}} = c(Y_{\text{ref}}), \quad (23)$$

with $c(\cdot)$ from Eq. (13) and the slope from Eq. (14). **The legacy constant is not a special case of this one.** At $Y_{\text{ref}} = 0.5$, Eq. (23) returns ≈ 6.07 , not 4, because a mid-*luminance* grey is code 0.735 where the transfer curve is steep.

Error 3, pushing the other way: retinal illuminance. $E = 500$ Td in Eq. (7) models a brighter-adapted and therefore *more sensitive* observer than the real ≈ 115 Td implied by $Y = 0.097$ at a 100 cd/m^2 peak white through Eqs. (16)–(17). Correcting it *reduces* CSF and therefore *raises* the admissible amplitude, partly cancelling errors 1 and 2.

The corrected budget, in closed form. Substituting Eq. (23) into Eq. (21) and converting to code units by dividing by the slope gives a budget in linear units and then in sRGB units:

$$B_{\text{lin}}(f) = \frac{\tau Y_{\text{ref}}}{2 \text{CSF}(u(f); Y_{\text{ref}})}, \quad (24)$$

$$B_{\text{sRGB}}(f) = \frac{B_{\text{lin}}(f)}{(dY/dc)|_{c_{\text{ref}}}} = \frac{\tau (c_{\text{ref}} + 0.055)}{4.8 \text{CSF}(u(f); Y_{\text{ref}})}. \quad (25)$$

The second equality is exact for $c_{\text{ref}} > 0.04045$: writing $v = (c_{\text{ref}} + 0.055)/1.055$, Eq. (12) gives $Y_{\text{ref}} = v^{2.4}$ and Eq. (14) gives $dY/dc = (2.4/1.055)v^{1.4}$, so

$$\frac{Y_{\text{ref}}}{dY/dc} = \frac{1.055 v^{2.4}}{2.4 v^{1.4}} = \frac{1.055 v}{2.4} = \frac{c_{\text{ref}} + 0.055}{2.4}, \quad (26)$$

and $2 \times 2.4 = 4.8$. Here $\text{CSF}(u; Y_{\text{ref}})$ denotes Eq. (7) evaluated with E set from Y_{ref} by Eqs. (16)–(17).

Equation (26) is itself a result and Sec. 10.3 depends on it: **the sRGB budget is linear in the code value, not in the luminance.** The 2.4 exponent cancels between the transfer function and its slope, so gamma encoding has already performed most of the luminance adaptation for free.

The measured size of the error. Over the full train split, legacy budget divided by calibrated budget:

	low (1.5 cpd)	mid (7.7 cpd)	near-Nyquist (37.5 cpd)
legacy / calibrated	2.98×	1.97×	1.88×

decomposing at Nyquist as $3.35\times$ from $\mu = 0.5$ alone and $0.56\times$ from $E = 500$ Td pushing back, for a net $1.88\times$. **Every τ quoted under the legacy convention describes a patch roughly twice as visible as claimed**, and roughly three times at low frequency. This is finding **F2**.

5 The residual: shape, scale, and what survives compositing

5.1 A smooth spectral reparameterisation, not a projection

Given an unconstrained parameter $\mathbf{z}$, the residual is built so that its spectrum respects B by construction [**no external source**]:

$$\mathbf{Z} = \mathcal{F}\mathbf{z}, \quad (27)$$

$$\hat{\mathbf{Z}}(f) = B(f) \frac{\mathbf{Z}(f)}{\sqrt{1 + |\mathbf{Z}(f)|^2}}, \quad (28)$$

$$\boldsymbol{\delta}_0 = \mathcal{F}^{-1}\hat{\mathbf{Z}}, \quad (29)$$

where $|\cdot|$ is complex modulus and the operations are elementwise over bins $f \in \mathcal{K}$. Since $|\mathbf{Z}|/\sqrt{1 + |\mathbf{Z}|^2} < 1$ for every finite $\mathbf{Z}$, we have $|\hat{\mathbf{Z}}(f)| < B(f)$ for every bin and every input — an unconditional bound.

Three properties are load-bearing:

- **It is a reparameterisation, not a projection.** A hard projection zeroes the gradient of every component sitting on its bound, which is exactly where an adversarial component wants to live. The squash in Eq. (28) is smooth everywhere, the bound is never attained exactly, and no component can become stuck.
- **Phase is preserved exactly.** The scaling multiplies the *complex* coefficient, so only magnitude is limited. This also avoids differentiating $\arg(\mathbf{Z})$, which is undefined at the origin.
- **It is scale-invariant in $\mathbf{z}$ up to shape.** Only the shape of the initialisation matters, which is why $\mathbf{z}^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ rather than $\mathbf{0}$: Eq. (32) below is $0/0$ at $\boldsymbol{\delta}_0 = \mathbf{0}$.

Because $\mathcal{F}^{-1}$ of a half-spectrum is real by construction, no conjugate-symmetry bookkeeping is required.

5.2 The visibility index

Per-bin compliance is not enough, because the visual system sums evidence across frequency and orientation channels: many individually sub-threshold components can be jointly visible. This is *probability summation*, standardly modelled by a Minkowski norm with exponent $\beta \approx 3\text{--}4$ (Quick, 1974; Robson and Graham, 1981; Watson, 1979). Define the per-bin visibility and the pooled index

$$v(f) = \kappa |\mathcal{F}\boldsymbol{\delta}(f)| \cdot \text{CSF}(u(f)), \quad (30)$$

$$V_\beta(\boldsymbol{\delta}) = \left(\sum_{\text{ch}=1}^3 \sum_{f \in \mathcal{K}} v_{\text{ch}}(f)^\beta \right)^{1/\beta}, \quad \beta = 3. \quad (31)$$

Here $v(f)$ is the contrast of bin f expressed in threshold units (so $v = 1$ is exactly at threshold), the inner sum runs over half-spectrum bins, the outer sum over the three colour channels — each channel is transformed separately — and $V_\beta \leq 1$ means at or below the detection threshold of the modelled observer *under the assumed geometry*.

Why not the max. $\max_f v(f)$ is the $\beta \rightarrow \infty$ limit and therefore the most permissive criterion available; defaulting to it would flatter every result. Concretely: a residual whose every bin respected its own budget measured a max-visibility of 0.235 while reaching a pixel-space deviation of ± 7.65 — nominally invisible and in fact catastrophic. Report V_β ; report $\max_f v(f)$ only as a secondary diagnostic.

The channel term is not cosmetic. Because the outer sum runs over three channels, a bound derived on one channel is $3^{1/\beta} = 1.44\times$ too generous. This mistake was made once and caught by measurement (Sec. 6.2).

5.3 Shape and scale are two separate operations

The full reparameterisation composes the envelope with a global rescale [**no external source**]:

$$\delta_1 = \delta_0 \cdot \frac{\tau}{V_\beta(\delta_0 \odot \mathcal{M})}, \quad (32)$$

where $\odot \mathcal{M}$ restricts to the silhouette actually pasted (for a square patch $\mathcal{M} \equiv 1$ and the restriction is vacuous), and the scale is computed *per sample* so that one image in a batch cannot borrow another’s headroom.

The division of labour is exact and worth stating in the thesis in these terms:

shape	Eq. (28): the 1/CSF envelope biases the residual’s energy toward the bands the eye discards.
scale	Eq. (32): one global factor sets total perceived visibility to exactly τ .

The two must be separate. The envelope alone bounds each bin but says nothing about their sum; the rescale alone would let the attack put energy wherever it likes, including where the eye is most sensitive. Normalising to *exactly* τ rather than clamping at it is deliberate: the attack always wants the largest perturbation it is allowed, so the constraint is active at every step anyway, and an equality is smooth where a min would introduce a non-differentiable kink at the bound.

Corollary — τ acts once, not twice. τ appears in both Eq. (21) and Eq. (32), but its occurrence in the budget is inert whenever $A_{\max}$ is inactive: $B \propto \tau$ implies $\delta_0 \propto \tau$ implies $V_\beta(\delta_0) \propto \tau$, and the ratio in Eq. (32) cancels it. Measured at $S = 128$ and the default geometry, varying τ in Eq. (21) alone over $[0.05, 1]$ leaves the residual bit-identical.

5.4 The range fit: why clipping cannot be used

The residual is added to a base $\mathbf{r}$ and the sum must lie in $[0, 1]$. A clamp is the obvious choice and is fatal. A clamp is a hard nonlinearity: it does not scale a residual, it *bends* it, and a bent peak is a corner, and a corner is broadband — energy lands in exactly the low and mid bands where CSF is largest and the budget tightest. Measured: a residual built to exactly 1.0 JND rose to 5.0 JND after clamping against a real image, with only 8% of pixels actually clipped.

A *uniform rescale* is the only operation that leaves the spectrum’s shape untouched — scaling δ by c scales every Fourier coefficient by c , so a CSF-shaped residual stays CSF-shaped and $V_\beta(c\delta) = c V_\beta(\delta)$ exactly. Define per-pixel headroom and the admissible ratio [**no external**

source]

$$h_i = \begin{cases} 1 - r_i, & \delta_i > 0 \quad (\text{pushing up}), \\ r_i, & \delta_i \leq 0 \quad (\text{pushing down}), \end{cases} \quad (33)$$

$$\gamma_i = \begin{cases} +\infty, & h_i \leq h_{\min} \text{ or } i \notin \text{supp}(\mathcal{M}), \\ \frac{h_i}{|\delta_i|}, & \text{otherwise,} \end{cases} \quad (34)$$

$$\delta_2 = \delta_1 \cdot \text{clip}(Q_q(\{\gamma_i\}), 0, 1), \quad (35)$$

where i indexes pixels and channels, r_i is the base value at i , Q_q is the empirical q -quantile with $q = 0.001$, and $h_{\min} = 10^{-3}$.

Two exclusions in Eq. (34) are each worth a paragraph in the thesis because each caused a silent total failure:

- **Saturated pixels are excluded, not down-weighted.** A pixel already at 0 or 1 has zero headroom in one direction, so its ratio is 0 and it drags any quantile to 0 with it. Over a 128×128 patch a 0.1% quantile tolerates ≈ 49 such pixels; over a full 512×1024 frame it tolerates 524, and a single Cityscapes sky blow-out exceeds that. The observed consequence was a full-image probe that scaled *every* band to exactly zero and produced no measurement. Excluding them is also physically correct: where the base is already 1.0 and the residual pushes up, the pixel was never going to move, so it should not veto the scale of every pixel that could.
- **The mask is not optional.** A shaped patch loads its cutout padded with transparent black, so for one reference 45% of the square is RGB 0 and 95% of the region outside the silhouette is. Unmasked, those never-pasted pixels drove the quantile to 0.0000 and scaled the residual to exactly zero — a five-epoch run that produced the unmodified reference image. The same quantile computed inside the silhouette was 1.26, i.e. the residual would have fitted at full scale with room to spare.

Allowing the lowest q of pixels to clip trades a tiny, measurable violation for a usable attack. The violation is not assumed away: it is measured by Sec. 5.5.

5.5 Enforcement: bounding what survives, not what was intended

Everything above constrains the residual we *intend* to add. The observer sees

$$\delta^{\text{real}} = \text{clip}(\mathbf{r} + \delta_2, 0, 1) - \mathbf{r}, \quad (36)$$

which is a different signal, and the difference is not small. The honest number to report is therefore the *realised visibility*

$$V^{\text{real}} = V_\beta(\delta^{\text{real}} \odot \mathcal{M}). \quad (37)$$

τ is an intent; V^{real} is an outcome; the gap between them is the measure of how well the constraint held.

The gap was large (finding F3). On SegFormer, image 420, $\tau = 0.25$, five seeds at 1000 steps: V^{real} came out $2.77 \times \tau$ on the $\mu = 0.5$ convention (worst seed $4.52\times$) and $7.44\times$ on the locally measured convention (worst $12.34\times$). The constraint held at initialisation ($\tau = 0.25$ requested, 0.2508 realised) and broke *during* optimisation, because the attack pushes toward the largest perturbation it is allowed and so drives more pixels into the clamp. The signature is diagnostic: the worst seed for visibility had the *lowest* residual RMS and absolute maximum of the five — backwards for “bigger is more visible”, and exactly right for clipping, which removes peaks while adding harmonics.

et al., 2017):

1. δ is *shared* across the whole dataset;
2. the CSF is evaluated at one *fixed* reference luminance, because a universal patch cannot look at the image it lands on.

It is the non-adaptive control for the per-image attack, and its value is as much in a legible null result as in a success.

The composite is

$$\mathbf{x}_{\text{win}}^{\text{adv}} = \text{clip}(\mathbf{x}_{\text{win}}^{01} + \delta, 0, 1), \quad (40)$$

where $\mathbf{x}_{\text{win}}^{01}$ is the footprint window of the image in $[0, 1]$ code values. Every image receives the *same residual on different content*, which is the whole point.

6.2 Hard projection, and why the envelope must be normalised

Here the constraint is a true projection rather than a squash, because the bound must hold as an *inequality on the parameter itself* for “project the parameter, not the render” to be checkable:

$$\hat{\mathbf{Z}}(f) = \mathbf{Z}(f) \cdot \min\left(1, \frac{B'(f)}{|\mathbf{Z}(f)|}\right), \quad (41)$$

$$\pi = \frac{1}{|\mathcal{K}|} \left| \left\{ f : |\mathbf{Z}(f)| > B'(f) > 0 \right\} \right|, \quad (42)$$

with π the fraction of live bins sitting at their bound. Phase is again untouched. The dead gradient at a saturated bin is real, so π is reported rather than discarded: a run whose π climbs toward 1 has stopped optimising and is only re-phasing.

Why B' and not B . The obvious construction — project onto the raw $1/\text{CSF}$ budget, then rescale the result so that pooled visibility equals τ — does not work. The projection clamps most bins down, so pooled visibility afterwards is far below τ , and the rescale that restores it multiplies every bin back up: a smoke test put bins at $90\times$ their own budget. Per-bin bound and pooled equality are not simultaneously satisfiable on one envelope.

Normalise the *envelope* instead [**no external source**]:

$$V_{\text{full}} = \left(3 \sum_{f \in \mathcal{K}} \left(\kappa B(f) \text{CSF}(u(f)) \right)^\beta \right)^{1/\beta}, \quad (43)$$

$$B'(f) = B(f) \cdot \frac{\tau}{V_{\text{full}}}, \quad (44)$$

where V_{full} is the pooled visibility of a hypothetical residual sitting at budget on *every* bin, and the factor 3 is the colour-channel count of Eq. (31). Then the per-bin bound is the real constraint and pooled visibility is an *inequality*, $V_\beta \leq \tau$, with equality only if every bin saturates — the honest direction for the inequality to point.

The factor 3 was worth measuring. Omitting it makes the envelope $3^{1/3} = 1.44\times$ too generous. Measured before the term was added: pooled 0.0541 against a nominal $\tau = 0.05$, an 8% overshoot that every assertion in the training loop passed, because the assertion and the normaliser shared the mistake.

Consequence for comparison. A universal run at nominal τ is *less* visible than a per-image run at the same nominal τ , where Eq. (32) enforces equality. The two must therefore be compared at matched *realised* visibility as well as at matched nominal τ .

6.3 Projection belongs in the optimiser step, not the forward pass

Placing Eq. (41) inside the forward pass makes the map radially flat above the bound, so $\partial\mathcal{L}/\partial|\mathbf{Z}|$ is *exactly* zero at any saturated bin and its magnitude can never come back down. Saturation becomes an absorbing state. Measured from the default initialisation: π sat at 0.989 for 200 Adam steps while the per-bin spend ratio moved by 2×10^{-5} — the optimiser could only rotate phase, and the spectral allocation this mode exists to measure was frozen at 1.000 by construction rather than chosen. Applying the projection as a post-step parameter clamp, with the loss seeing an unflattened gradient, fixes it.

6.4 Geometry is pinned

The universal mode refuses to run unless $S = p$. Resampling a residual resamples its spectrum, so the budget its bins were projected onto would stop describing the pasted signal, and τ would silently cease to mean anything. At $H = 512$, $s = 0.25$ that is $S = p = 128$. The same warning applies to the per-image mode whenever `-patch_scale` and `-patch_size` are changed independently.

6.5 Calibrated visibility: the headline metric for this mode

τ is set once at a fixed Y_{ref} . What that one residual actually *costs* on each image is

$$Y_i = \frac{\sum_{(u,v) \in \mathcal{M}} Y(\mathbf{x}_{i,uv}^{01})}{|\mathcal{M}|}, \quad (45)$$

$$\tau_i^{\text{cal}} = V_\beta(\boldsymbol{\delta} \odot \mathcal{M}; \text{CSF}(\cdot; Y_i), \kappa_{\text{cal}}(Y_i)), \quad (46)$$

with $Y(\cdot)$ from Eq. (15), $\text{CSF}(\cdot; Y_i)$ from Eq. (7) with E set by Eqs. (16)–(17), and κ_{cal} from Eq. (23). *Both* the sensitivity and the contrast denominator respond to Y_i , so this is not a rescale of the nominal number and the per-image ordering can differ from it.

The sentence this exists to license is: *“the budget was set at $\tau = a$ using a fixed reference; realised visibility across the val set ranged from b to c .”* That spread is the empirical case for content dependence, or the evidence against it.

6.6 Clipping is tracked, not assumed away

Equation (40) truncates where content is near 0 or 1, so the realised perturbation is not the projected one and τ is violated in the permissive direction. Report

$$\text{frac}_{\text{clip}} = \frac{1}{3|\mathcal{M}|} \left| \left\{ i \in \mathcal{M} \times \{1, 2, 3\} : x_i + \delta_i \notin [0, 1] \right\} \right| \quad (47)$$

every step. A non-negligible value invalidates the guarantee quietly, which is the worst way for it to fail. The alternative composite (`-csf_composite_fit`, Eq. (35)) rescales rather than truncates and so preserves the spectrum — at the cost of a per-image scale, which is precisely the per-image adaptation a universal patch is not supposed to have. Clipping is the default for that reason. Note also that `clip` has zero gradient on the saturated side, so a heavily clipped run also loses gradient on those pixels.

7 Application, threat model, and the optimisation loop

7.1 Compositing

Following the localised-patch threat model of Brown et al. (2017), with optional silhouette mask after Tan et al. (2021),

$$\tilde{\mathbf{P}} = (\text{Resize}_{p \times p}(\mathbf{P}) - \boldsymbol{\mu}_n) \oslash \boldsymbol{\sigma}_n, \quad (48)$$

$$\mathbf{m} = \text{Pad}_{t,\ell}(\mathbb{K}_{p \times p} \odot \mathcal{M}), \quad (49)$$

$$\mathbf{x}^{\text{adv}} = \mathbf{x} \odot (1 - \mathbf{m}) + \text{Pad}_{t,\ell}(\tilde{\mathbf{P}}) \odot \mathbf{m}, \quad (50)$$

where $\mathbf{P}$ is the rendered patch in $[0, 1]$ from Eq. (39), $\oslash$ is elementwise division, (t, ℓ) is the top-left placement, and $\text{Pad}_{t,\ell}$ zero-pads the patch into the full frame at that offset. The footprint follows the *silhouette*, not the bounding box, so with a cutout only genuinely occluded pixels are excluded from scoring — stricter and more honest than excluding the whole square.

Placement. Three modes: `center`; `semantic`, which reads the clean prediction; and `gradcam`, which reads a sensitivity map. The last two impose an ordering constraint — clean forward, then resolve placement, then apply — and getting it backwards silently falls back to centre, so that every downstream number is quietly about a different patch position.

7.2 Optimisation

$$\mathbf{z}^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \mathbf{z}^{(j+1)} = \text{Adam}(\mathbf{z}^{(j)}, \nabla_{\mathbf{z}} \mathcal{L}), \quad j = 0, \dots, K-1, \quad (51)$$

with $\mathcal{L}$ one of the objectives of Sec. 8, K the step count, and Adam (Kingma and Ba, 2015) on the raw gradient. The network f is frozen throughout and $\nabla_{\phi} f$ is never formed. The learning rate follows a cosine schedule (Loshchilov and Hutter, 2017).

The learning rate is not transferable between modes, and this is finding F6. The two modes optimise different objects:

	per-image (csf)	universal (universal_csf)
what $\mathbf{z}$ is	a <i>direction</i> : mapped through a scale-invariant squash and renormalised to exactly τ	<i>the residual itself</i> , in pixel units (RMS 0.027 at $\tau = 0.25$)
what lr controls	how fast that direction rotates	movement inside a small ball; overshoot is discarded by the projection
correct lr	0.2	0.01

Scaling the per-image parameter by $1000\times$ moves the rendered patch by 7.5%. Measured mean relative change of the applied residual per step:

lr	csf	universal_csf
0.2	0.50	1.89
0.05	0.084	1.09
0.01	0.031	0.15
0.001	0.0038	0.039

A relative change above 1 means each step discards the previous residual entirely. Corroborating diagnostic: π after 200 steps is 0.931 at lr 0.2 (pinned to the boundary, phase-only) against 0.508 at lr 0.01 (healthy interior). **Watch π in the logs:** pinned above ≈ 0.9 means the learning rate is too large whatever any script says.

Annealing is not optional at these rates. A flat learning rate was measured swinging 9.6 mIoU points across four *identical* single-image invocations, because Adam’s step is $\approx \text{lr}$ per coordinate whatever the gradient is, so the run never settles and the reported number is wherever the walk happened to stop.

8 The three untargeted objectives

All three objectives below are *untargeted*: they specify no class the attack should impose, only that the prediction should be wrong or uncertain. They are the axis on which the loss comparison is run. The targeted objective (IPatch-style, driving every scored pixel toward one class Mirsky, 2023) is a separate arm and is not part of this axis; it is mentioned only where the contrast is informative.

8.1 The scored support, shared by all three

Every objective is a mean over a pixel set Ω , never over the whole frame:

$$\Omega = \underbrace{\{(u, v) : \mathbf{y}(u, v) \neq 255\}}_{\text{valid labels}} \setminus \underbrace{\text{supp}(\mathbf{m})}_{\text{patch footprint}} \cap \underbrace{\mathcal{R}}_{\text{reach mask (optional)}}. \quad (52)$$

Footprint exclusion is not optional. Leave the patch region in the loss and the cheapest way to reduce it is to make the *patch* look wrong, which collapses the objective while the remote effect stays exactly zero (Mirsky, 2023). The attack we are studying is a *remote* one; scoring the footprint measures occlusion.

Reach restriction, and what it can and cannot buy. A patch can only influence pixels inside the model’s effective receptive field (Luo et al., 2016). Measured with a label-free random-stimulus probe, prediction-change collapses below 1% past ≈ 300 px at 512×1024 and ≈ 150 px at 256×512 ; of $\approx 450\text{k}$ remote pixels only $\approx 50\text{--}80\text{k}$ are reachable. The unreachable $\approx 85\%$ contribute gradients that are small but not zero, and that do not correspond to achievable flips: they dilute the mean by $\approx 5\times$ and add variance to the update direction. Restricting Ω to $\mathcal{R}$ therefore (a) shrinks the denominator and (b) removes those contributions outright. **It cannot extend reach** — the reachable set is fixed by the architecture. Available as a radial disc of default radius $0.62H$ (set from the measured collapse point, not chosen) or as the measured mask itself.

Reporting asymmetry, to be declared. The loss support and the evaluation metric cover different pixel sets: we optimise on $\mathcal{R}$ but still report the drop over *all* remote pixels. This is deliberate — the metric must not be tuned to the method — and must be stated wherever a reach-restricted number appears.

The guarded mean. Every objective reduces with

$$\langle g \rangle_{\Omega} = \frac{\sum_{(u,v)} g_{uv} \mathbb{I}[(u, v) \in \Omega]}{\max(|\Omega|, 1)}, \quad (53)$$

where g_{uv} is the per-pixel quantity. The clamped denominator is not defensive programming for its own sake: Cityscapes batches occur in which void regions (ego-vehicle hood, borders) plus the patch footprint cover every pixel, giving 0/0. The resulting NaN propagates through the backward pass, permanently poisons Adam’s moment estimates, and freezes the patch at its initial value for the rest of training with no error raised. That cost one 100-epoch run before it was noticed.

8.2 Objective 1 — cross-entropy

The simplest untargeted objective maximises the network’s cross-entropy against the ground truth:

$$\text{CE}_{uv} = -\log p_{uv, \mathbf{y}(u,v)}, \quad (54)$$

$$\mathcal{L}_{\text{CE}} = -\langle \text{CE}_{uv} \rangle_{\Omega} = \langle \log p_{uv, \mathbf{y}(u,v)} \rangle_{\Omega}. \quad (55)$$

Here $p_{uv,k}$ is the softmax probability of class k at pixel (u, v) from Sec. 2, and $\mathbf{y}(u, v)$ is that pixel’s ground-truth class. Cross-entropy is *high* when the prediction is wrong, so the attack maximises it; since Adam minimises, the returned scalar carries a leading minus sign. This is the segmentation analogue of the untargeted FGSM/PGD objective (Goodfellow et al., 2015; Madry et al., 2018).

Its known pathology, which motivates the other two. $-\log p_y$ is unbounded above, so once a pixel is thoroughly misclassified it can still contribute a large and growing gradient. The mean is then dominated by a few already-won pixels, and the attack keeps spending capacity where it has already succeeded instead of on the pixels still holding out. Gu et al. (2022) make the same observation for segmentation PGD and address it by reweighting correctly- and incorrectly-classified pixels over the attack schedule.

8.3 Objective 2 — CosPGD

CosPGD (Agnihotri et al., 2024) rescales the per-pixel cross-entropy by the cosine similarity between the predicted distribution and the one-hot ground truth:

$$w_{uv} = \cos(\mathbf{p}_{uv}, \mathbf{e}_{\mathbf{y}(u,v)}) = \frac{\langle \mathbf{p}_{uv}, \mathbf{e}_{\mathbf{y}(u,v)} \rangle}{\|\mathbf{p}_{uv}\|_2 \|\mathbf{e}_{\mathbf{y}(u,v)}\|_2} = \frac{p_{uv, \mathbf{y}(u,v)}}{\|\mathbf{p}_{uv}\|_2}, \quad (56)$$

$$\mathcal{L}_{\text{CosPGD}} = -\langle [w_{uv}]_{\text{sg}} \cdot \text{CE}_{uv} \rangle_{\Omega}, \quad (57)$$

where $\mathbf{e}_k$ is the one-hot vector of class k (so $\|\mathbf{e}_k\|_2 = 1$, giving the third equality), and $[\cdot]_{\text{sg}}$ denotes a stop-gradient.

The cosine is a weight, never the objective. $w_{uv} \in (0, 1]$ measures how aligned the prediction still is with the truth. Pixels still predicted correctly ($w \approx 1$) keep full weight; pixels already fooled ($w \approx 0$) are de-emphasised. The gradient itself comes from CE, which does not saturate the way a raw cosine does near alignment. This is the direct answer to Sec. 8.2’s pathology: effort is concentrated on the pixels that have not yet flipped.

Two declared deviations from the paper.

1. **The cosine is detached.** Agnihotri et al. (2024) do not detach it; in their Algorithm 1 the gradient flows through the whole product. Detaching removes gradient contributions through the weight and is a deliberate deviation, common in patch-attack implementations.
2. **Adam replaces sign-SGD with ε -projection.** CosPGD is a PGD variant (Madry et al., 2018), and a patch is not an ε -ball perturbation, so Eq. (57) is optimised with Adam on the raw gradient.

Both are correct for this threat model and both must be stated in the thesis, because a reader comparing numbers against the CosPGD paper is otherwise comparing two different algorithms.

8.4 Objective 3 — Tsallis entropy

Status: specified here, not yet implemented. This subsection is the implementation contract as well as the derivation. The entropy itself is Tsallis’s (1988) one-parameter generalisation of Shannon entropy (Shannon, 1948), independently introduced by Havrda and Charvát (1967); its use as an *adversarial objective for segmentation* carries no external source here and is marked [no external source].

Definition. For a distribution $\mathbf{p} \in \Delta^{C-1}$ and an entropic index $q > 0$, $q \neq 1$,

$$S_q(\mathbf{p}) = \frac{1}{q-1} \left(1 - \sum_{c=1}^C p_c^q \right). \quad (58)$$

q is the single hyper-parameter; C is the class count; p_c is the probability of class c . $S_q \geq 0$ with equality if and only if $\mathbf{p}$ is one-hot, and S_q is maximised by the uniform distribution $p_c = 1/C$, where it attains the q -logarithm of C :

$$S_q^{\max} = \ln_q C = \frac{C^{1-q} - 1}{1 - q}. \quad (59)$$

For $C = 19$ this is 6.718 at $q = 0.5$, $\ln 19 = 2.944$ in the Shannon limit, and 0.947 at $q = 2$ — the range varies by an order of magnitude with q , which is why the objective must be normalised before different q are compared:

$$\bar{S}_q(\mathbf{p}) = \frac{S_q(\mathbf{p})}{\ln_q C} \in [0, 1]. \quad (60)$$

The objective.

$$\mathcal{L}_{\text{Tsallis}} = -\langle \bar{S}_q(\mathbf{p}_{uv}) \rangle_{\Omega}, \quad (61)$$

with Ω from Eq. (52) and $\langle \cdot \rangle_{\Omega}$ from Eq. (53). Minimising $\mathcal{L}_{\text{Tsallis}}$ maximises the mean normalised entropy of the remote prediction: the attack drives the network toward *indecision* rather than toward a specific error. It is the sign-flipped counterpart of entropy *minimisation* as used in semi-supervised learning and test-time adaptation (Grandvalet and Bengio, 2004; Wang et al., 2021).

The Shannon limit. Writing $q = 1 + \epsilon$ and expanding $p_c^{1+\epsilon} = p_c e^{\epsilon \ln p_c} = p_c(1 + \epsilon \ln p_c) + O(\epsilon^2)$,

$$S_{1+\epsilon}(\mathbf{p}) = \frac{1}{\epsilon} \left(1 - \sum_c p_c - \epsilon \sum_c p_c \ln p_c \right) + O(\epsilon) \xrightarrow{\epsilon \rightarrow 0} - \sum_c p_c \ln p_c = H(\mathbf{p}), \quad (62)$$

using $\sum_c p_c = 1$. So $q \rightarrow 1$ recovers Shannon entropy exactly, and Eq. (60) recovers the normalised entropy $H(\mathbf{p})/\ln C$ that the diagnostics already report. **An implementation must dispatch $q = 1$ to the Shannon form**, since Eq. (58) divides by $q - 1$.

Gradient, in closed form. Let $\mathbf{z}_{uv} \in \mathbb{R}^C$ be the logits at a pixel and $\mathbf{p} = \text{softmax}(\mathbf{z})$. From Eq. (58), $\partial S_q / \partial p_c = -q p_c^{q-1} / (q - 1)$, and the softmax Jacobian is $\partial p_c / \partial z_k = p_c(\delta_{ck} - p_k)$ with δ_{ck} the Kronecker delta. Chaining,

$$\begin{aligned} \frac{\partial S_q}{\partial z_k} &= \sum_c \frac{-q p_c^{q-1}}{q-1} p_c(\delta_{ck} - p_k) = \frac{-q}{q-1} \left(p_k^q - p_k \sum_c p_c^q \right) \\ &= \frac{q}{q-1} p_k \left(\sum_c p_c^q - p_k^{q-1} \right). \end{aligned} \quad (63)$$

Reading it: a logit is pushed *up* when its class is less probable than the “typical” class under the q -weighting $\sum_c p_c^q$, and *down* when it is more probable. The objective is therefore a *levelling* force, and it needs no label to compute — which is the whole point.

Sanity check against the Shannon limit. Substituting $q = 1 + \epsilon$ into Eq. (63) and using $\sum_c p_c^{1+\epsilon} \approx 1 + \epsilon \sum_c p_c \ln p_c$ and $p_k^\epsilon \approx 1 + \epsilon \ln p_k$ gives $\partial H / \partial z_k = -p_k (\ln p_k + H(\mathbf{p}))$, the standard Shannon-entropy gradient. The closed form is therefore consistent at the boundary.

What q controls, and why it is worth sweeping. The factor p_k^{q-1} in Eq. (63) sets which part of the distribution the objective attends to.

$q > 1$	Emphasises the <i>dominant</i> class: p_c^q suppresses the tail, so the force is concentrated on pulling the winner down. Acts like an attack on confidence. As $q \rightarrow \infty$, S_q is governed by $\max_c p_c$ alone.
$q \rightarrow 1$	Shannon entropy: every class contributes in proportion to $p_c \ln p_c$.
$q < 1$	Emphasises the <i>tail</i> : p_c^q inflates small probabilities, so the force pushes probability mass out into many classes. As $q \rightarrow 0^+$, $S_q \rightarrow C - 1$ for any full-support $\mathbf{p}$ and the objective degenerates into counting the support.

q is therefore a second realism-independent knob and a genuine axis: $q \in \{0.5, 1, 2\}$ is the minimal informative sweep, run at fixed τ .

Properties that make it worth having as a third objective.

- **Label-free.** Equation (61) never references $\mathbf{y}$. It can therefore be scored on void pixels, it cannot be accused of using ground truth a real attacker would not have, and it is the natural objective for the universal patch of Sec. 6, where no per-image label is available at attack time.
- **Bounded.** $\bar{S}_q \in [0, 1]$, so unlike Eq. (55) no pixel can dominate the mean without bound. This is the same benefit CosPGD obtains by reweighting, but obtained by construction.
- **It targets a different failure mode.** CE and CosPGD produce *confident errors*; a maximum-entropy attack produces *abstention*. For a downstream consumer that thresholds on confidence, these are not the same attack at all, and the distinction is worth a paragraph in the discussion.

Two cautions that must be stated with any Tsallis result.

1. **Maximum entropy is not maximum damage.** The uniform distribution is the global optimum of Eq. (61), and while reaching it implies an arg max change in general, entropy can also rise substantially *without* any arg max changing. Never quote an entropy objective's success by its own loss value; quote $\Delta \text{mIoU}_{\text{rem}}$ and ϕ (Sec. 9), exactly as for the other two.
2. **Objective/metric coupling.** The diagnostic suite already reports normalised Shannon entropy by distance ring. That diagnostic is *the objective* for $q \rightarrow 1$, so it stops being independent evidence for a Tsallis run and becomes a training curve. Report it, label it as such, and carry the mechanism claim on the confusion flows and the reach curve instead.

Numerical requirements for the implementation. p_c^{q-1} diverges as $p_c \rightarrow 0$ for $q < 1$, so clamp $p_c \geq \epsilon = 10^{-12}$ before exponentiating; compute p_c^q as $\exp(q \log p_c)$ on the clamped values for stability; dispatch $q = 1$ to the Shannon branch; and normalise by $\ln_q C$ with the *same* C used to build Ω . Reuse Eq. (53) verbatim rather than `reduction="mean"`, for the reason given in Sec. 8.1.

8.5 What the three objectives are for

	CE	CosPGD	Tsallis
uses labels	yes	yes	no
per-pixel range	unbounded	unbounded, reweighted	bounded in $[0, 1]$
effort allocation	drifts to won pixels	held on unflipped pixels	held on confident pixels
failure mode induced	confident error	confident error	abstention
role in the thesis	baseline	main objective	label-free control and a different threat

The finding that applies to all three (F7). An untargeted objective has no class goal, yet it converges on *one* confusion channel. Measured: $\approx 97\%$ of all remote flips went road $\rightarrow$ car, with car IoU dropping ≈ 68 points in every image. The untargeted loss is an *implicit class selector*, not a broad disruptor — it finds the most exploitable boundary and exploits only that. Aggregate mIoU hides this completely; the source $\rightarrow$ destination flow table is what makes it visible, and a single flow carrying more than half of all flips is the criterion for calling a result *structured* rather than *diffuse*. These have different security implications and must not be reported interchangeably.

Expected contrast with a targeted objective. Under a targeted loss every scored pixel is pushed toward the *same* class, so per-pixel gradients align and add. Untargeted losses have pixel *A* saying “not road” and pixel *B* saying “not sky”, and the two partly cancel. That is the mechanistic reason the untargeted family converges on one channel anyway — and it also explains the sink-class variance of Sec. 10.4.

9 What to measure

9.1 Remote mIoU drop — the headline

$$\Delta \text{mIoU}_{\text{rem}} = \text{mIoU}(\mathcal{U}f(\mathbf{x}), \mathbf{y}; \Omega) - \text{mIoU}(\mathcal{U}f(\mathbf{x}^{\text{adv}}), \mathbf{y}; \Omega), \quad (64)$$

with the class mean taken over classes *present in* $\mathbf{y}|_{\Omega}$, so that both terms share a denominator (Cordts et al., 2016). Counting a class merely because it is *predicted* makes the two means run over different class sets, and a rare class entering or leaving the denominator shifts ΔmIoU by roughly $\text{mIoU}/(n-1)$ independently of any pixel-level change. This was an actual bug and it is worth one sentence in the methods section.

9.2 Flip rate

$$\phi = \frac{|\{(u, v) \in \Omega : \arg \max_k \mathcal{U}f(\mathbf{x})_{k,uv} \neq \arg \max_k \mathcal{U}f(\mathbf{x}^{\text{adv}})_{k,uv}\}|}{|\Omega|}. \quad (65)$$

ϕ is label-free, so it is the right instrument for the reach and frequency probes, which must not reference any class.

9.3 Influence beyond effect: margin and entropy by ring

The winner margin at a pixel is $\text{top}_1 - \text{top}_2$ of the logits. Its change under attack shows pixels the patch has pushed *toward* a flip without completing one; that region reliably extends past the flip zone, so the attack’s *influence* is wider than its *effect*. Normalised entropy $H(\mathbf{p})/\ln C$ complements it: the margin measures the top-2 gap, entropy measures uncertainty across all classes, and the two can disagree. Both are reported over fixed absolute distance rings (50 px steps to 1200 px) so that ring k is the same physical annulus in every image.

9.4 Spectral efficiency — the metric F1 requires

Prediction change per unit of perceptual cost, per radial band b :

$$\eta(b) = \frac{\phi(b) - \phi_0}{V_\beta(b)}, \quad (66)$$

where $\phi(b)$ is the flip rate induced by band-limited noise confined to b , ϕ_0 is the unattacked baseline, and $V_\beta(b)$ is that stimulus’s visibility from Eq. (31). **Quote η , not ϕ :** Sec. 10.1 is a statement about η and is not supported as a statement about ϕ .

9.5 Where the energy went

Two radial summaries, and they answer different questions.

$$\text{RAPSD}(b) = \frac{1}{|A_b|} \sum_{f \in A_b} |\mathcal{F}\delta(f)|^2, \quad (67)$$

$$\bar{a}(b) = \frac{1}{|A_b|} \sum_{f \in A_b} |\mathcal{F}\delta(f)|, \quad \text{spend}(b) = \frac{\bar{a}(b)}{\bar{B}(b)}, \quad (68)$$

with A_b the set of bins whose radial frequency falls in bin b . Equation (67) is *power* (Ruzanski and Chandrasekar, 2011) and is the right thing to compare against the natural-image power law (Field, 1987). Equation (68) is *amplitude*, because the budget is an amplitude bound and the quantity worth plotting is the ratio spent/allowed — meaningful only if both sides are in the same units.

Bounded by CSF does not imply statistically inconspicuous. Natural spectra follow an approximate power law (Field, 1987; Falck et al., 2025; Dieleman, 2024), so energy placed near Nyquist is *atypical of the data* precisely where the eye is least sensitive. Invisibility to a human and indistinguishability from natural image statistics are distinct properties; only the former is constrained here, and Eq. (67) is what makes the latter measurable. A detector trained on RAPSD is the obvious defence and should be named as such.

9.6 Uncertainty

Report a mean with an interval and n , never a single run. Population runs use a normal-approximation interval $\bar{x} \pm 1.96 \hat{\sigma} / \sqrt{n}$, chosen over a bootstrap (Efron and Tibshirani, 1993) only because at n in the hundreds they agree to well inside the width that matters and an interval a reader can recompute from $(\bar{x}, \hat{\sigma}, n)$ by hand is worth more than a slightly tighter one. $n = 1$ must return *no* interval rather than a zero-width one: a zero-width interval reads as “measured exactly”, which is the opposite of what one sample establishes. Pooled rates are pooled ratios — numerator and denominator summed over images, divided once — with the per-image spread reported alongside, never instead.

10 Findings

10.1 F1 — The network is spectrally flat; the gap is efficiency, not effectiveness

Measured on SegFormer MiT-B0, Cityscapes 512×1024 , 5 images, patch region, with band-limited noise injected at fixed RMS = 0.02 (the *equal-amplitude* control, not the visibility-normalised run):

band (cyc/px)	flip rate, remote	perceptual cost V_β
0.00–0.02	0.36%	70.7 JND
0.20–0.35	0.53%	5.5 JND
0.35–0.50	0.39%	1.5 JND

Every band moves the network about equally; their perceptual prices differ by a factor of ≈ 47 . Two corollaries, each a result in its own right:

- **The stride-4 patch embedding does not kill the high band.** That was the specific architectural risk this attack family was exposed to, and it did not materialise. This is the spectral counterpart to the ERF work: the ERF bounds reach *spatially*, and no comparable spectral bound was found.
- **The asymmetry lives entirely on the perceptual side.** The CSF constraint does not reduce what the attack can do; it reallocates a fixed perceptual budget to where it buys most. The original framing — “the network reads what the eye discards” — is half right: the network reads everything, and *that* is what makes the CSF weighting useful.

What F1 does not support. Flip-rate differences between bands (0.36–0.53%) sit inside their own between-probe spread (± 0.32 –0.63), so they must not be reported as a band ranking. Scene-dependence of the readable band is likewise *unsupported*: per-image peak bands came out [3, 4, 2, 4, 3] from means of 0.36–0.53% against standard deviations of 0.32–0.63, and the tool now refuses that verdict when band separation is below the between-probe spread. The hypothesis remains attractive — it would explain why a 150-epoch shared generator recovered ≈ 0 while per-image optimisation reached +16.31 — but it needs more probes and images before it can be claimed. The visibility-normalised runs ($\tau \in \{0.25, 0.5, 1, 2\}$) were all *inconclusive* against the probe’s 0.5% absolute floor (maxima 0.036/0.108/0.241/0.425%); their band profile nevertheless rose monotonically toward Nyquist in every τ and every scene, $\approx 40\times$ bottom to top. Report both the floor verdict and the profile consistency.

Relation to the low-frequency attack literature. A body of work argues that adversarial perturbations restricted to *low* frequencies transfer better and survive input transformations more robustly (Guo et al., 2020; Sharma et al., 2019). That is not in tension with F1, and the distinction is worth stating explicitly because a reader will otherwise assume one of the two must be wrong. Those results optimise for *transferability and robustness* under an L_p budget, where low frequencies are favoured because they survive resizing, JPEG and model change. This work optimises for *invisibility* under a *perceptual* budget, where low frequencies are the most expensive part of the spectrum by a factor of ≈ 47 (F1) and the L_p norm is not the constraint at all. The two literatures agree that the spectrum is the right place to think and disagree only about which end of it a given budget makes attractive.

10.2 F2 — The budget was $\approx 1.9\times$ too permissive

Derived in Sec. 4.4 and measured over the full train split ($n = 2975$). The practical instruction: report **both** a nominal and a calibrated τ column in every results table, so past runs are restatable by a constant factor rather than invalidated. The enforced τ deliberately keeps the legacy convention so the existing ladder stays comparable.

10.3 F3 — The constraint broke during optimisation, and enforcement fixes it

Sec. 5.5. Unenforced, realised visibility reached $2.77 \times \tau$ (worst seed $4.52\times$) on the legacy convention and $7.44\times$ (worst $12.34\times$) on the locally measured one, at 1000 steps; at 400 steps it

was $1.57\times / 3.78\times$, i.e. the violation *grows with run length*. Enforced runs hold 0.2500 exactly at every step. **Instruction: -csf_enforce realised for any number quoted with a τ attached.**

The efficiency reading, which is the interesting part. Drop per unit of *realised* visibility was better at 400 steps (best 143.7) than at 1000 (best 113.3). The extra 600 steps buy drop by spending perceptual budget, not by finding a cheaper attack. Any claim that longer optimisation improves the attack must therefore be made per unit of realised visibility, not per step.

10.4 F4 — Run length, and a bimodality that was not real

Same configuration, five seeds per cell:

steps	$\Delta\text{mIoU}_{\text{rem}}$	ϕ
400	35.60 ± 4.78	$63.1 \pm 6.7\%$
1000	50.53 ± 1.54	$98.95 \pm 0.33\%$
1000, enforced	45.95 ± 3.57	—

At 400 steps the distribution looked bimodal (three runs ≈ 39 , two ≈ 30 , nothing between). It was not two basins — it was truncation: every run was still descending when the step counter ran out. The run-to-run instability that made two identical configurations look like a code regression was mostly *too few steps*; a flat learning rate (F6) made it worse by never settling. Enforcement costs 4.58 points.

Where the variance went. At 1000 steps the seeds agree on drop (relative sd 3.1%) and disagree hard on cost (visibility sd 37.9%). Under enforcement the residual variance is a *sink-class choice*, not noise: a **building** sink cannot flip building, which is 28.1% of that frame, so its flip rate is capped at 71.9% (observed 68.66%, drop 40.86), whereas a **train** sink — a class absent from the image — flips building too and reaches 99.4% (the other four seeds: 47.22 ± 2.48). Forcing the sink with a targeted objective is the way to remove that lottery, and it is the cleanest available control for the implicit-class-selector finding (F7).

10.5 F5 — A fixed reference luminance costs almost nothing

Making the CSF budget’s reference luminance per-image buys only $\approx 13\%$ of budget at Nyquist. Over the full train split ($n = 2975$), p95/p5 budget ratio in sRGB code units:

	low	mid	near-Nyquist
train, centre (barten)	$1.63\times$	$1.16\times$	$1.13\times$
train, road (0.75, 0.5)	$1.67\times$	$1.16\times$	$1.13\times$
val, centre (barten)	$1.68\times$	$1.17\times$	$1.14\times$

Worst case over the *full* range (min to max Y , a $93\times$ span in linear luminance) is still only $1.39\times$ at Nyquist. Val tracks train (median Y 0.103 vs 0.097), so an L_{ref} fitted on train is valid for val. Placement does not dominate: the road surface is 18% darker in median than centre but only slightly wider in spread.

Mechanism — and it is not “Barten is flat”. Equation (26): the sRGB budget is proportional to $c_{\text{ref}} + 0.055$, so gamma encoding has already performed most of the luminance adaptation, the 2.4 exponent cancelling against the same exponent in the transfer slope. Barten’s sensitivity then pushes back a *further* 0.59–0.85 $\times$, because a darker background is also a less sensitive observer. Measuring in linear luminance instead reports 7–13 $\times$ and is measuring in a

space nothing in the attack lives in. The SSO control confirms the decomposition independently: having no luminance parameter, its ratio is flat at $1.930\times$ across all frequencies, exactly $4.85\times$ (luminance) $\times 0.398\times$ (sRGB slope). Barten’s frequency-dependent pushback is the entire difference between 1.93 and 1.13.

Scope — read before quoting this. This measures exactly *one* term: the reference luminance the budget is evaluated at. The per-image mode is content-adaptive in several other ways, none of which this touches:

mechanism	per-image today	measured by F5?
base = the region the patch covers	yes	no
range fit, Eq. (35) (per-image scale)	yes	no
the optimised residual itself	yes	no
placement (semantic / gradcam)	yes	no
CSF budget reference luminance	no (fixed $\mu = 0.5$)	yes , and it buys $\approx 13\%$

Never write “content-adaptivity has no perceptual justification” — that over-reads this measurement by a wide margin. Argue for per-image adaptivity on the residual, base, range fit and placement, and on attack strength; do not argue for it on the grounds that a fixed reference luminance mis-states visibility, because it barely does.

10.6 F6 — The parameterisation determines the learning rate

Sec. 7.2. `csf` takes lr 0.2, `universal_csf` takes lr 0.01, and carrying one across to the other silently destroys the run in one direction and freezes it in the other. Annealing is mandatory: a flat learning rate swung 9.6 mIoU points across four identical invocations. Monitor π (Eq. (42)) and $\text{frac}_{\text{clip}}$ (Eq. (47)) as first-class diagnostics, not as debug output.

10.7 F7 — The untargeted loss is an implicit class selector

Sec. 8.5. $\approx 97\%$ of remote flips on one channel, road $\rightarrow$ car, with car IoU down ≈ 68 points in every image. Report the flow table alongside every mIoU number, and classify the result as *structured* (one flow $> 50\%$) or *diffuse*.

10.8 F8 — Reach is bounded spatially, and the bound is architectural

Prediction-change from a maximal random stimulus collapses below 1% past ≈ 300 px at 512×1024 and ≈ 150 px at 256×512 ; $\approx 50\text{--}80\text{k}$ of $\approx 450\text{k}$ remote pixels are reachable. This is label-free and target-independent by construction, which is what makes it a valid control for the geometric factor rather than a correlate of the attack. Restricting the loss support to the reachable set shrinks the denominator $\approx 5\times$; it cannot extend reach, and a null result there is itself a test of the two-factor model.

10.9 F9 — Hard manifold constraints fail; soft and spectral ones do not

Constraining the patch to a 120-dimensional BigGAN latent — a hard manifold constraint — found nothing adversarial inside it: 0.71% flip rate against 99.63% for the unconstrained parameterisation on SegFormer, across two architectures and both objectives. A 150-epoch shared conditional generator likewise recovered ≈ 0 where per-image optimisation reached +16.31. The spectral constraint of Sec. 5 is the opposite kind of object: it keeps the full pixel parameterisation and reshapes rather than restricts it, which is why it works where the latent constraint did not. This is the argument for why the universal mode of Sec. 6 is the *right* control — it isolates universality from parameterisation.

10.10 The τ ladder, and where it stops meaning anything

Below the knee of Eq. (35), realised visibility tracks τ and the residual scales linearly:

τ	0.05	0.10	0.25	0.50
residual RMS	0.0049	0.0098	0.0245	0.0490

Above the knee the range fit binds, attained visibility saturates, and τ ceases to control anything. The knee is reference-dependent — it is set by the headroom of $\mathbf{r}$ — and lay near $\tau \approx 0.4$ for the references used here. **Usable range:** $\tau \in [0.05, 0.5]$. Values above the knee must not be reported as though achieved.

11 Experimental design

11.1 The reference configuration

Everything below varies one axis against this fixed point. It is the configuration the single-image result was promoted from, and it is chosen rather than defaulted:

setting	value	why this value
architecture	SegFormer MiT-B0	the model every finding above was measured on (Xie et al., 2021)
resolution	512×1024	reach and band profiles are resolution-dependent (F8)
patch	$S = p = 128, s = 0.25$	pinned; $S \neq p$ resamples the spectrum and invalidates τ (Sec. 6.4)
placement	center	keeps the footprint comparable across images; gradcam is a separate rung, not a default
mode	csf -from_image	the base is the covered content (Sec. 5.6)
τ	0.25	the rung every existing number is quoted at
enforcement	-csf_enforce realised	mandatory for any number quoted with a τ (F3)
steps	1000	400 truncates mid-descent and fakes bimodality (F4)
lr	0.2, cosine	correct for csf only ; flat swings 9.6 points (F6)
objective	CosPGD	the main objective; CE and Tsallis are the axis
log interval	25 steps	40 points for the convergence band, and the resolution at which “best” is checkpointed

11.2 The axes worth running, in order

#	axis	what it settles
1	τ ladder $\{0.05, 0.1, 0.25, 0.5\}$	The deliverable is the <i>curve</i> , not any operating point: strength against perceptibility, the role α plays in Tan et al. (2021). Stop at 0.5; above the knee τ controls nothing.
2	objective $\{\text{CE}, \text{CosPGD}, \text{Tsallis}\}$	Sec. 8. Hold τ , placement, steps and lr fixed. Tsallis additionally sweeps $q \in \{0.5, 1, 2\}$ at fixed τ .
3	per-image vs universal	The adaptivity control (Sec. 6). Compare at matched <i>nominal</i> τ and at matched <i>realised</i> visibility — the two modes enforce τ as an equality and an inequality respectively, so nominal matching alone is not a fair comparison.
4	CSF-bounded vs unconstrained ceiling	A drop under a perceptual constraint is only interpretable next to the unconstrained ceiling <i>on the same images</i> .
5	CSF model $\{\text{barten}, \text{sso}\}$	Shows the visibility claim does not depend on one CSF, and isolates the luminance decomposition (Sec. 3.3, F5).
6	geometry sweep	Every visibility number is conditional on θ . Report at least the desk-monitor default and the physical-patch geometry of Eq. (5).
7	architecture	F1 and F8 are single-architecture results. The frequency probe is label-free and dataset-independent, so it transfers to ADE20K weights and can bracket several backbones cheaply.

11.3 Sample size, pairing, and resumption

One image is an anecdote. Neither a drop nor a realised-visibility number survives being quoted from a single image, and neither survives being quoted from a single seed (Sec. 10.4). Run the identical configuration over a population using the *same* function the single-image path uses, so that the two cannot drift.

$n = 100$, growing by permutation. $n = 100$ supports a ± 2 point claim at the spread this attack has shown; $n = 500$ is the whole Cityscapes val set. The sample is a *seeded permutation truncated to n* , so $n = 200$ is $n = 100$ plus 100 new images and a resumed run attacks only the new ones. Changing the seed instead throws the first 100 away. Report the observed spread and the n it would take to support the claim being made, and raise n only if that number says so.

Pair by image index. The constrained arm and the ceiling arm must use the same sample seed, and the comparison must be the *paired* difference. Given this attack’s scene-to-scene spread, the paired test is the only one that resolves a small effect; an unpaired comparison of two means at $n = 100$ will not.

Pilot before committing walltime. Five images first, so the walltime is arithmetic rather than a guess — 1000 steps is $3.3\times$ the length the population loop was previously run at.

11.4 What every run must report

A run that does not report these is not quotable.

1. $\Delta \text{mIoU}_{\text{rem}}$ with an interval and n (Eq. (64)); never a single number.
2. Realised visibility V^{real} (Eq. (37)), *both* nominal and calibrated (F2), as a distribution and not a mean.
3. The geometry θ in deg/px, and the Nyquist frequency it implies.
4. π (Eq. (42)) for projected modes and $\text{frac}_{\text{clip}}$ (Eq. (47)) for clipped composites.
5. The convergence band: loss and drop against step, with an explicit verdict on whether the tail is flat. A run whose tail still climbs is truncated, not converged (F4).
6. The confusion flow table and the structured/diffuse verdict (F7).
7. The spend ratio by band, Eq. (68), against the budget it was allowed.
8. The seed count and the per-seed spread, for single-image results.

11.5 Claim discipline

The measurements above license some sentences and forbid others. This table is the one to keep next to the writing.

Supported	Not supported
“Per unit of perceptual cost, near-Nyquist perturbation is $\approx 47\times$ more efficient than low-frequency perturbation” (F1).	“The network reads frequencies the eye cannot see.” It reads all of them; the difference is price.
“The readable band is a model property, consistent across the scenes tested” — with the inconclusive verdict stated.	“The readable band is scene-dependent.” The separation is inside the between-probe spread (F1).
“ τ nominal = 0.25; calibrated, that is ≈ 0.47 ” (F2).	Any τ quoted without saying which convention it is in.
“Under enforcement the constraint holds at τ exactly, at a cost of 4.6 mIoU points” (F3).	“The constraint holds by construction.” It did not; it was violated $2.8\text{--}7.4\times$ before enforcement.
“A fixed reference luminance costs $\approx 13\%$ of budget at Nyquist” (F5).	“Content adaptivity has no perceptual justification.” F5 measures one term only.
“ $\approx 97\%$ of flips ran on one channel; the untargeted objective acts as an implicit class selector” (F7).	“The untargeted attack causes broad-spectrum confusion.”
“Below the modelled detection threshold of an average observer under the stated geometry.”	“Imperceptible.” That requires a study with human observers, which none of this constitutes.

11.6 Sequencing, and the cheap results first

Three of the most load-bearing measurements need no training and no GPU time, and should be re-run whenever a constant changes: the footprint luminance survey (F2, F5), the frequency-sensitivity probe (F1, label-free so it transfers across weights), and the ERF probe (F8). Run those before any ladder; a premise that fails there fails cheaply.

12 Provenance, verification, and threats to validity

12.1 Provenance

Sec.	Element	Source
3.1	cycles/pixel $\rightarrow$ cycles/degree	Galinska et al. (2026, App. D)
3.1	screen and physical-patch geometry constructors	[no external source]
3.2	Barten CSF; parameter values	Barten (2003); params Galinska et al. (2026, App. A)
3.3	Standard Spatial Observer, oblique effect	Watson and Ahumada (2005); Watson (2000)
3.4	sRGB transfer function and its slope	International Electrotechnical Commission (1999)
3.4	luminance weights	ITU-R (2015)
3.4	pupil diameter, trolands	Barten (1999, Ch. 2)
4.1	Michelson contrast, $\kappa = 2/\mu$	Michelson (1927)
4.2	<i>inversion of CSF into an amplitude budget</i>	[no external source]
4.3	<i>patch-specific low-frequency cutoff</i>	[no external source]
4.4	<i>code-value calibration of κ and the closed-form budget</i>	[no external source]
5.1	<i>smooth spectral reparameterisation</i>	[no external source]
5.2	Minkowski pooling / probability summation	Quick (1974); Robson and Graham (1981); Watson (1979)
5.4	<i>uniform-rescale range fit</i>	[no external source]
5.5	<i>bisection enforcement on the composited residual</i>	[no external source]
5.6	<i>covered content as the residual base</i>	[no external source]
6.1	universal perturbation framing	Moosavi-Dezfooli et al. (2017)
6.2	<i>envelope normalisation to pooled τ</i>	[no external source]
6.5	<i>calibrated per-image visibility</i>	[no external source]
7.1	localised patch application	Brown et al. (2017)
7.1	silhouette masking	Tan et al. (2021)
7.2	Adam; cosine annealing	Kingma and Ba (2015); Loshchilov and Hutter (2017)
8.1	footprint exclusion	Mirsky (2023)
8.1	effective receptive field	Luo et al. (2016)
8.2	untargeted CE attack lineage	Goodfellow et al. (2015); Madry et al. (2018); Xie et al. (2017)
8.2	the “effort drifts to won pixels” pathology	Gu et al. (2022)
8.3	CosPGD objective; declared deviations	Agnihotri et al. (2024); deviations [no external source]
8.4	Tsallis entropy	Tsallis (1988); Havrda and Charvát (1967)
8.4	<i>Tsallis entropy as an untargeted segmentation attack</i>	[no external source]
8.4	entropy objectives, opposite sign	Grandvalet and Bengio (2004); Wang et al. (2021)
9.1	mIoU protocol, Cityscapes	Cordts et al. (2016)
9.5	RAPSD	Ruzanski and Chandrasekar (2011)
9.5	natural-image spectral statistics	Field (1987); Falck et al. (2025); Dieleman (2024)
9.6	bootstrap, as the alternative not taken	Efron and Tibshirani (1993)
—	LPIPS, evaluation only, never in the objective	Zhang et al. (2018)
—	physical-world threat model, EOT, printability	Athalye et al. (2018); Sharif et al. (2016)

Twelve elements carry no external source and are contributions of this work. From Galinska et al. (2026) this work takes a function, its constants, a unit conversion and two scoping caveats — not a method: their contribution is a weighting over flow-matching timesteps, which has no analogue here. Their α , which interpolates $w_{\text{final}} = \alpha w_{\text{CSFlow}} + (1 - \alpha)w_{\text{base}}$, is *not* τ : it blends two weightings and bounds nothing.

12.2 Bibliography status

Entries were assembled from the verified bibliography of `docs/csf_patch_procedure.tex` plus standard references, and carry a DOI or an unambiguous venue wherever one exists. Three notes:

- **galinska2026csflow** — the arXiv identifier is carried over unchanged from the companion document and has not been independently re-checked here. Verify it once before submission, since it is cited for the CSF parameters, the unit conversion, and two caveats.
- **gu2022segpgd** — SegPGD is ECCV **2022**. A comment in `validation.sh` attributes it to ECCV 2024; that is a typo worth fixing at the source so the wrong year does not propagate into the thesis.
- **Tsallis as an attack objective** — Tsallis (1988) and Havrda and Charvát (1967) are cited for the *entropy*. No source is claimed for its use as an adversarial segmentation objective; it is marked **[no external source]** in Sec. 12.1. If a prior use is found in the literature, the marking must change, and that is a search worth doing before the loss chapter is written.

12.3 Threats to validity

1. **τ is nominal, not a psychophysical JND.** Three approximations stand between them, each biasing it: the mean luminance in κ is assumed rather than measured (quantified, F2); the Minkowski sum in Eq. (31) runs over DFT bins rather than perceptual channels; and Eq. (2) fixes a viewing geometry that Galinska et al. (2026, Sec. 6) identify as needing calibration. τ should be reported as a relative axis.
2. **No human study.** Nothing here establishes imperceptibility. The most that Eq. (6) supports is a consistency check by eye at the matched viewing distance.
3. **Single architecture, single dataset.** F1 and F8 are SegFormer results on Cityscapes. The probes that produce them are label-free and therefore cheap to replicate elsewhere; until they are, the claims are architecture-scoped.
4. **Digital threat model.** There is no expectation over transformations (Athalye et al., 2018) and no printability term (Sharif et al., 2016) in the objective. A physical patch would need both, and the geometry of Eq. (5) is the right starting point — it is more favourable than the desk-monitor default, not less.
5. **Invisibility is not inconspicuousness.** The constraint bounds visibility to a modelled human observer. It does not bound detectability by a spectral classifier, and by placing energy near Nyquist it makes the patch *atypical* of natural image statistics (Sec. 9.5). Name this defence rather than waiting for a reviewer to.
6. **The linearisation in Eq. (14)** is first-order and holds for small residuals. At large τ , trust the measured realised visibility rather than the budget.

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